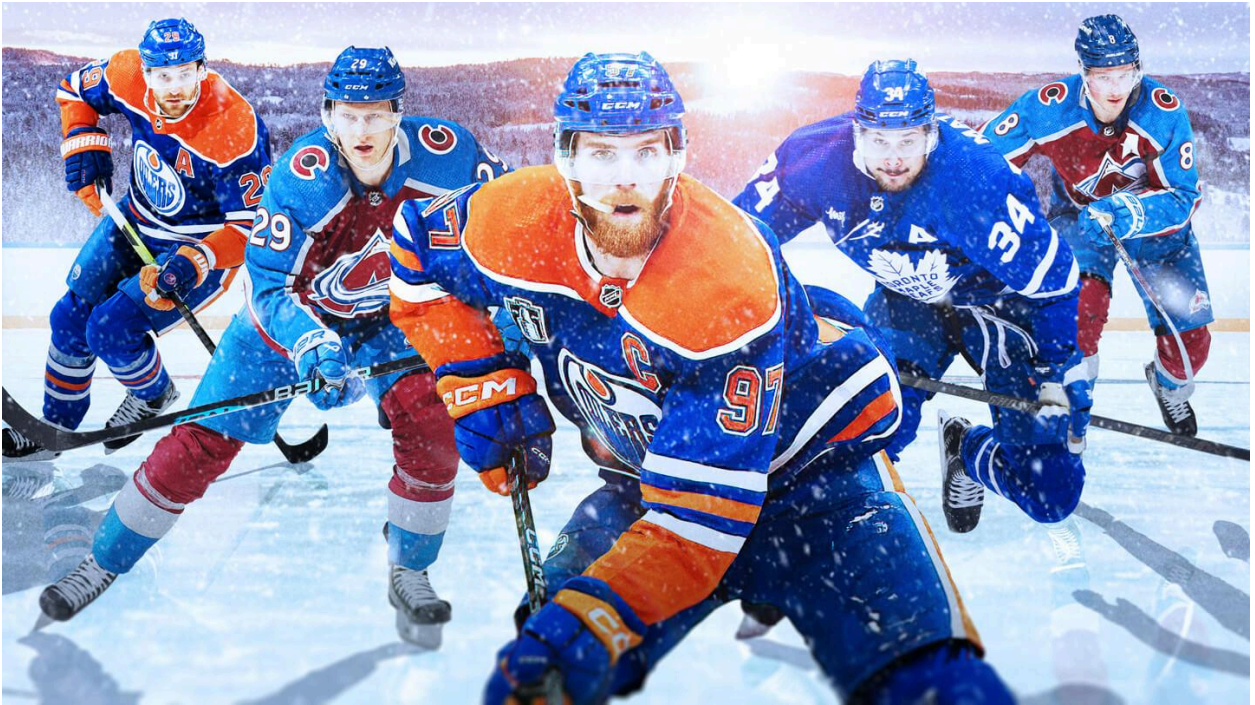


Do Higher Paid NHL Players Perform Better?



Introduction

The goal of my project is to identify if players who are performing are being compensated appropriately. Will we find outliers who are underpaid with a lot of production? Maybe players who are paid too much with little production? Using a public dataset that compares salary to point production, I plan to use google colab to clean the data set, summarize the data, and visualize what is happening when it comes to the salary/point correlation

Data Selection

I chose a publicly available csv from [Kaggle.com](https://www.kaggle.com). It has hundreds of rows of data reflecting player salaries and point production through different stats. I chose this dataset because I coach hockey and my brother is also a professional hockey player. This data interests me knowing how productive my brother is in his league.

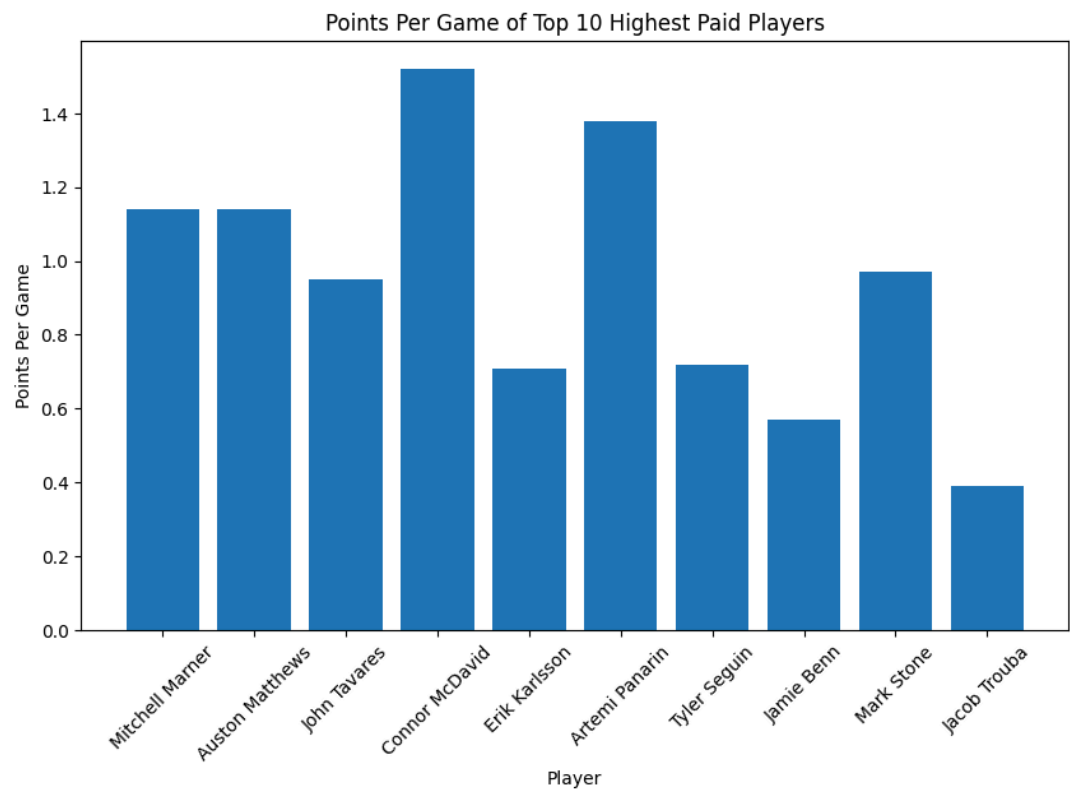
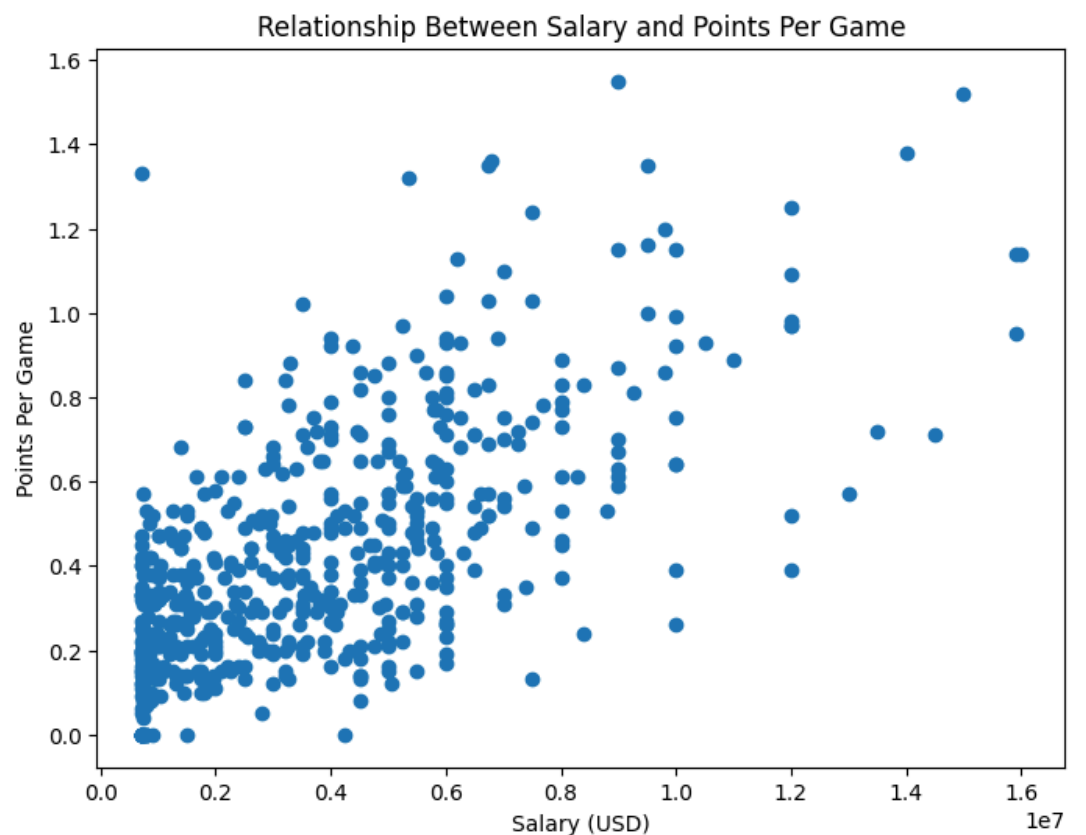
Data Analysis

I started by searching for duplicate and missing values. This dataset is clean reporting 0 duplicated values. It does however show 425 missing clause values. Not all players have contract clauses and this value isn't important to our research so, I will drop it from our set. The dataset is now clean and ready for exploratory data analysis.

Upon analyzing the data, we found that the average player had 24.5 points in a season with a median of 20. The most points recorded was 110. Breaking down points into goals and assists, the average goals scored is 9.2, with the highest goal scorer recording 48. Assists had an average of 15.3 with a maximum of 67. Another important value to interpret is games played. The average number of games played in the season is 51.5 with the most being 72. Overall, we should note games played influences total point production. Games played carries values from 1 all the way to 72, because of this let's interpret point per game averages with salaries.

This is where the data gets fun! The data shows a strong positive correlation of 0.67 between salary and performance. It generally means higher paid players score more points per game. However, the relationship isn't perfect. Some higher paid players underperform whereas some lower paid players overperform. Ultimately, the results show that salary and performance are strongly related, however, player outcomes do still vary.

Data Visualization



Conclusion

Overall, the dataset demonstrates that many NHL players are compensated according to production. Higher paid players typically have high point per game averages. There are always outliers in datasets and this one is no exception. Data can't tell us why some guys are paid higher because of their locker room presence or leadership abilities despite lack of scoring. It can't tell us when highly effective guys are team players who take lower salaries to help their team win a Stanley Cup. There are a lot of moving pieces in hockey that cannot be visually represented. They are like the unwritten rules. We know some longtime captains will offer up their salary when the cap is tight. We know some rookies will way out perform their rookie salary cap. What the data does show, is that hockey is a sport with a higher positive correlation between point production and salary.

References

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- Sonny Sachdeva . (n.d.). *THE NHL'S TOP 50 PLAYERS: 10-1*. Sportnet.ca. Photograph. Retrieved from <https://www.sportsnet.ca/nhl/longform/sportsnet-insiders-rank-the-nhls-top-50-players-10-1/>.